

your favourite MCU, as long as it meets all your application requirements, and the toolset that you already have.

Wi-Fi. Wi-Fi is one of the most common technologies for wireless connectivity, providing secure, reliable and fast Internet connectivity with IP protocol. Wi-Fi is awesome because it is available all over the place.

For high-capacity and high-quality mobile real-time applications that require video and voice capability, 802.11n provides operational advantages, with higher data throughput, greater range and robust link quality.

802.11ac offers twice the bandwidth of 802.11n. Use of multiple antennae (MIMO technology) further increases data rate and range.

nRF24L01 wireless transceivers with a 2.4GHz antenna provide easy implementation. The modules claim to support a communication range of 80 metres. With this range, you can play with your robot around the house or use it indoors to explore all rooms.

Diymall Esp8266 serial Wi-Fi wireless transceiver module Esp-01 takes care of Wi-Fi and TCP/IP stack. It supports SPI and UART connections. Both connections are ideal for pushing data up to the Internet for the Internet of Things (IoT) applications.

ESP8266 Esp-12e claims to offer a complete solution for Wi-Fi networking. It features storage capabilities and can choose intelligently and dynamically the best network available.

ZigBee. Zigbee/Xbee modules are advanced RF transceivers operating in the licence-free 2.4GHz frequency band. Zigbee transceivers are used where robust RF communication with high noise immunity and guaranteed packet delivery is required. The most common application of Zigbee modules is in sensor networks.

Digi ZigBee modules are long-range high-speed serial wireless

TABLE II
WI-FI MODULES AND DEVELOPMENT BOARDS

Company	Description
 Atmel	Wi-Fi / 802.11 modules SmartConnect ATWINC1500B-MU-T module
 Microchip	Onboard TCP/IP stack, ultra-low power, UART interface, programmable GPIOs and ADCs
 AdaFruit	Wi-Fi / 802.11 development tools Adafruit Feather Huzzah with ESP8266 Wi-Fi
 Seeed Studio	Wi-Fi / 802.11 development tools NodeMCU v2 - Lua based ESP8266 development kit

communication modules that can give a range of 30 metres indoors or 100 metres outdoors. This module is ideal for robot-to-robot or robot-to-PC communications.

Microchip's MRF24J40MA ZigBee module comes with an SPI interface with 120 metres range.

TARANG ZigBee modules manufactured by Melange systems can be interfaced using UART interface and can provide RF range of 100 metres to 50 kilometres with a directional antenna.

Thread. Thread is a new open global wireless standard developed to balance and improve on the increasingly important requirements of reliability, security, power efficiency and cost effectiveness. It runs on IEEE 802.15.4 physical radio specification and operates in the unlicensed ISM bands including 2.4GHz (v1.0).

Silicon Labs EM358x and EM359x families are optimised for both ZigBee and Thread networking stacks.

Proprietary. For applications that require longer range or support for proprietary protocols such as home automation, security and smart metering, sub-GHz and 2.4GHz propri-

etary wireless implementations offer substantive benefits over standards based, feature-rich protocols such as Wi-Fi, Bluetooth and ZigBee.

Flex Gecko is a family of 2.4GHz wireless SoCs targeted at proprietary wireless market.

Anurag Ambedkar, vice president - international business, Maven Systems Pvt Ltd, says, "The most important aspect of buying a wireless communication module has to be the end solution. It is necessary to consider various factors like range, cost and deployment geography. While protocols like ZigBee are successful in North America and Europe, the same does not stand true for Asia. As a result, the preferred modules should be by a company that is experienced in the local market and one that can customise to suit local needs."

He adds, "Maven has developed a wide variety of Wi-Fi modules using chipsets of Atheros, Broadcom, Roving Networks, AzureWave and Microchip. These modules communicate on either 2.4GHz or 5GHz frequency, along with AES, TKIP or WEP encryption for security. Maven has also developed a Wi-Fi-to-USB dongle that is being used by the US defence services."

Ambedkar further adds, "For RF modules, Maven's WiART series is used in a wide variety of domains like smart metering, smart lighting and medical wearables. We support a wide variety of frequencies including 865MHz-868MHz, 915MHz and 2.4GHz. These modules are protocol-agnostic and work with minimum change to the existing device protocol. Protocols supported by WiART include ZigBee protocol, BLE and Maven's proprietary Sub1 GHz protocol (SmartMesh)."

Akash Agrawal, chief executive officer, InkOcean, informs, "At InkOcean, I have sold a large variety of modules, some of which include Microchip's IoT module RN2483, which is a low-powered